

Algorithms

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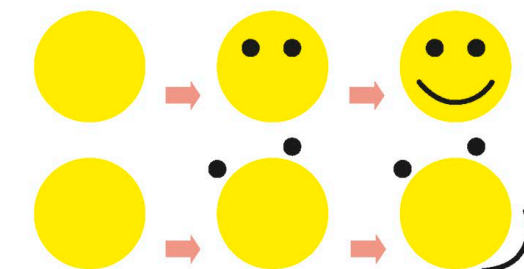
Though the word itself might sound unfamiliar, we all use algorithms every day. Baking a cake, knitting a jumper, or putting together a piece of furniture are all activities that use algorithms.

What is an algorithm?

An algorithm is a series of steps to solve a problem or carry out a task. To develop an algorithm, start by using decomposition to break down the problem into smaller tasks, then look for patterns in these tasks, and finally ignore unimportant details. This should give you the information you need to create an algorithm made of small steps that can all be described very clearly.

▽ Step by step

Algorithms describe a series of steps that must happen in sequence in order for the problem to be solved. In athletics, the triple jump competition involves the competitor running, then performing a hop, a bounce, and a horizontal jump at specific places in order to record a successful effort.



△ Unambiguous

Each step in an algorithm must be precise and unambiguous, with only one possible meaning. Vague instructions give incorrect results. An algorithm for drawing a smiley face might read: “draw a circle, then a curved line, and then two dots”. But this doesn’t tell us where the curved line and the dots should go in relation to the circle, or each other.

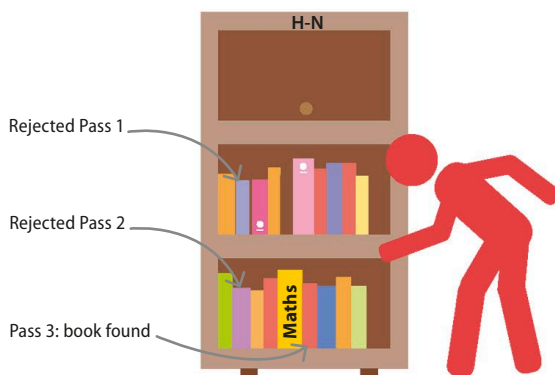
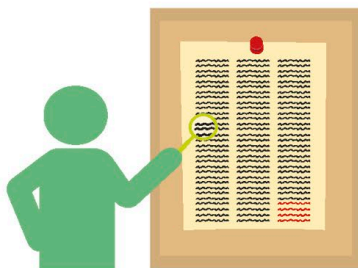


Types of algorithm

Algorithms exist for many different computer tasks: from smartphone apps that can tell what song is being played to the algorithms used by online search engines. One area where algorithms are very influential is data-processing; in particular, algorithms for searching and sorting data. There are different kinds of searching and sorting algorithm.

▷ Linear search

To find one item in a million, start at the first one and see if it’s the right item. If it is, stop searching; otherwise, look at the next item. This isn’t efficient as it might involve looking at every item on the list.



△ Binary search

For data that’s already sorted – for instance, an alphabetical bookshelf – a binary search is efficient. At each stage, you decide which half of the data the item you want is in. The half you don’t need is discarded. This is repeated until the item is found.